

CANCER DIAGNOSIS: BASELINE ANALYSIS

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1 ANALYSIS (4 PAGES)

[2 points] This section should include plots. For example, how key metrics vary with a specific hyperparameter, task complexity, etc.

1.1 **[2 points]** INTRINSIC METRICS

These are not the task itself (they might overlap with auxiliary losses) but are skills the model should have.

Intrinsic Metric 1: similarity structure. Metric 1 compares intra-grade similarity and inter-grade similarity in the learned representation space. A positive separation gap means that patients with the same Nottingham grade are more similar to one another than patients from different grades. In addition to the aggregate intra/inter statistics, we visualize a grade-to-grade similarity heatmap, which provides a finer-grained view of how each pair of grades aligns in the learned representation space.

XGB: the representation is the 3-dimensional class-probability vector. XGB produces an average cosine similarity of 0.8808 within Grade 1, 0.7652 within Grade 2, and 0.7569 within Grade 3, while the overall inter-grade similarity is 0.7430. This indicates that all three grades are, on average, more internally similar than the cross-grade baseline, with the clearest separation appearing for Grade 1. At the aggregate level, XGB yields an intra-class similarity mean of 0.7745 and a positive separation gap of 0.0315, suggesting weak but meaningful grade-related structure.

Basic Multimodal MLP: This is a basic MLP based on concatenated image and non-image features. The overall inter-grade similarity is 0.3397 and this is substantially lower than the intra-grade similarity results. Overall intra-grade was 0.4123 for Grade 1, 0.3655 for Grade 2, and 0.3772 for Grade 3. Patients do not have strong similarity based on grade in this test but more similarity was shown within Grade 1 when compared to Grade 2 and Grade 3.

Basic Multimodal Swin + MLP: The model uses Swin transformer as the image encoder and min-max scales the tabular data of clinical features. It uses two project layers to project two modalities into the same dimension and concatenates the two embeddings. The average cosine similarity for grade 1, grade 2, and grade 3 are 0.784, 0.782, and 0.788 respectively. The average inter-cross-grade similarity is 0.785. Overall, the cosine similarity of embeddings of the same grade is higher than those of different grades.

HetMed + N4 + RGB: the fused representation produces an average cosine similarity of 0.1595 within Grade 1, 0.1719 within Grade 2, and 0.3538 within Grade 3, while the overall inter-grade similarity is 0.2247. This indicates that only Grade 3 forms a noticeably tighter group in the current embedding space, whereas Grades 1 and 2 are not more internally similar than the average cross-grade relationship. At the aggregate level, the model yields an intra-class similarity mean of 0.2187 and a slightly negative separation gap of -0.0060, showing that the current multimodal representation still does not separate grades as cleanly as the XGB baseline.

HetMed + N4 + Grayscale: the version using a representation based on N4 correction and replication of 3 grayscale images produces an average cosine similarity of 0.64 within Grade 1, an average cosine similarity of 0.54 within Grade 2, and an average cosine similarity of 0.42 within Grade 3. The overall inter-grade similarity is 0.51. This indicates that based on the embedding space, patients

* Everyone Contributed Equally – Alphabetical order

within Grade 1 and patients within Grade 2 are more similar to each other than the similarity of patients that does not consider a specific grade. Patients within Grade 3 had the lowest similarity.

SimCLR + Grayscale: Analysis of the Resnet18-SimCLR representation reveals that Grade 1 possesses the highest internal cohesion with an intra-grade similarity of 0.401, while also showing the strongest cross-grade relationships, particularly with Grade 2 (0.444). The inter-grade similarity between Grade 1 and Grade 2 (0.444) actually exceeds their respective internal similarities (0.401 and 0.343), suggesting a significant spatial overlap between these two groups in the embedding space. In contrast, Grade 3 appears more isolated and distinct, exhibiting the lowest overall similarities with Grade 1 (0.365) and Grade 2 (0.261).

HetMed + RGB + Swin: this model is an extension of the Swin baseline. The model uses the concatenated embedding from the Swin baseline to construct the multiplex graph neural network. The average cosine similarity for grade 1, grade 2, and grade3 are 0.927, 0.919, and 0.915 respectively. The average inter-grade similarity is 0.906. It indicates that the consensus embeddings generated by the multiplex graph neural network lack discriminative power for tumor grade classification.

Intrinsic Metric 2: clustering structure. Metric 2 evaluates whether the learned representation induces clusters that align with Nottingham grades. We report ARI, NMI, and purity, and visualize the learned structure using a t-SNE projection in which color denotes cluster assignment and marker shape denotes the true Nottingham grade. For visualization, we first cluster the embeddings with $k = 3$ and then project them into 2D, which makes the cluster/grade relationship easier to inspect.

XGB: achieves ARI 0.0325, NMI 0.0741, and purity 0.5189. These values are modest, but they are still stronger than those of the current multimodal result. In the t-SNE cluster plot, the XGB representation shows weak but visible local grouping structure, which is consistent with the slightly positive Metric 1 separation results.

Basic Multimodal MLP: resulted in ARI 0.0298, NMI 0.0484, and purity 0.4972. These t-sne results show that the clusters are only slightly better than random grouping of the true grades.

Basic Multimodal Swin + MLP: the metrics show that even though the clusters are compact, there is little shared information within a cluster and the clustering is not better than random guessing.

HetMed + N4 + RGB: achieves ARI -0.0099, NMI 0.0101, and purity 0.4939. The t-SNE cluster plot still shows substantial grade overlap even when local neighborhoods are made more visually explicit, which helps explain why the clustering structure remains weak even after the graph redesign and longer training run.

HetMed + RGB + Swin: the model achieves slightly better clustering with higher intra-cluster similarities. However, clustering is still not better than random guessing.

HetMed + N4 + RGB: the model yields clusters with slightly better shared information within a cluster. However, the clustering is still not much better than random guessing.

HetMed + N4 + Grayscale: resulted in ARI 0.0761, NMI 0.0815, and purity 0.522. The t-sne results show that each of the clusters contain one grade more than the others, but the clusters do not correspond strongly to grade.

SimCLR + Grayscale: achieves ARI of 0.0066, NMI of 0.033, and purity of 0.59. These results show that although each cluster contains a primary grade, the overall clusters do not correspond strongly to the true grades.

Intrinsic Metric 3: variance ratio Metric 3 measures the variance contribution of each modality during fusion. For both image and tabular data modality, the variance contribution should be around 50%.

HetMed + N4 + RGB: Metric 3 estimates modality contribution by comparing the variance of pre-fusion image embeddings and pre-fusion tabular features. The current result is strongly imbalanced: the image variance fraction is 0.9620, while the tabular variance fraction is 0.0380. Therefore, the fused representation is still dominated by the image branch.

HetMed + N4 + Grayscale: Comparing the impact based on normalized variance using image embeddings and tabular features shows a substantial difference. The non-image tabular feature

variance ratio is 0.7024 and the image feature variance ratio is 0.2976. In this data the multimodal representation is dominated by the non-image features.

XGB: there is no multimodal fusion, so Metric 3 is not a variance-decomposition analysis. Instead, XGB functions as a no-fusion, non-image reference point for interpreting the multimodal result.

Basic Multimodal MLP: A comparison based on variance using image embeddings and tabular features shows that around three quarters of the modality variance is from the image data. The non-image tabular feature normalized variance ratio is 0.2475 and the image feature normalized variance ratio is 0.7525. When compared to other strategies like the HetMed + N4 + Grayscale version of the model, the contribution of variance from image and non-image features in the embedding space shows an opposite pattern.

Basic Multimodal Swin + MLP: Since the architecture of the model is designed to eliminate dominance of a single modality, the difference of variance contribution between two modalities is smaller compared to other models. However, the image features still carry more variations and might dominate the learning.

HetMed + RGB + Swin: the model is an extension of the multimodal baseline model with Swin and uses the fused embeddings generated by the baseline model. As a result, the variance analysis is the same.

XGB (TABULAR-ONLY BASELINE)

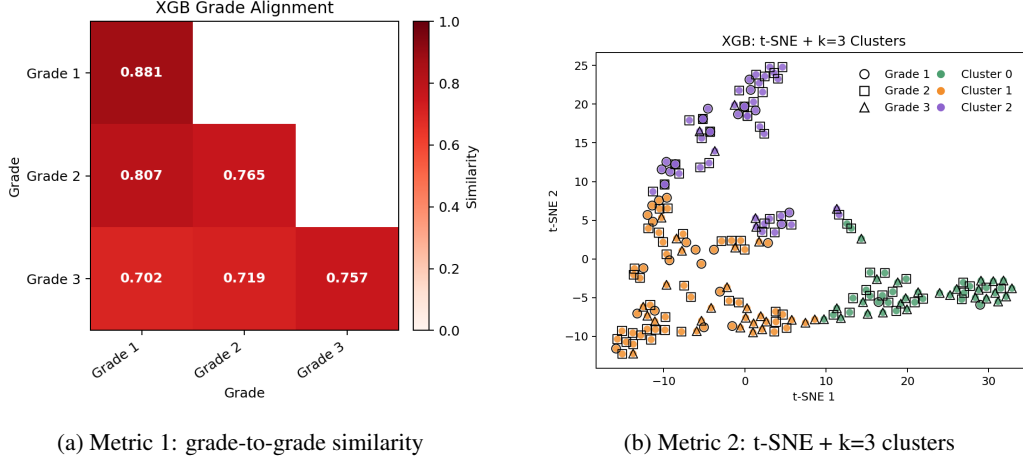


Figure 1: Intrinsic analysis for the XGB baseline. We use the grade-to-grade similarity heatmap as the main visualization for Metric 1, and a t-SNE cluster plot for Metric 2. This subsection can be expanded later if additional XGB variants are added by the team.

BASIC MULTIMODAL MLP (TABULAR DATA AND IMAGE DATA BASELINE)

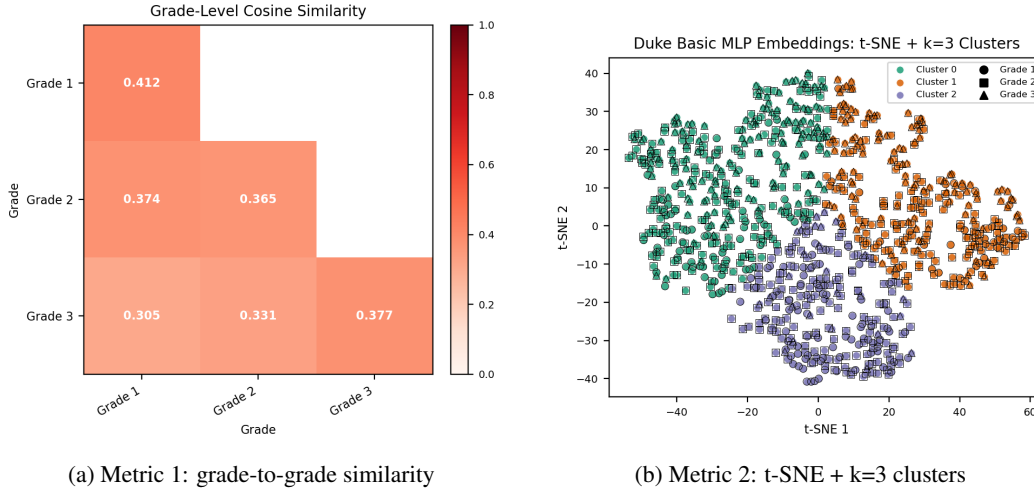


Figure 2: Intrinsic analysis for the Basic Multimodal MLP baseline.

BASIC MULTIMODAL SWIN + MLP(TABULAR DATA AND IMAGE DATA BASELINE)

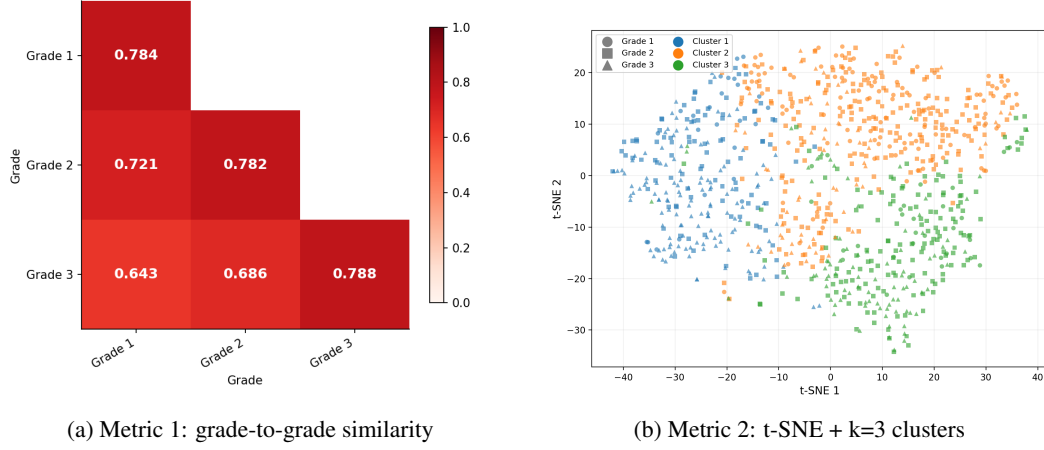


Figure 3: Intrinsic analysis for the Basic Multimodal MLP baseline with Swin transformer.

HETMED + N4 + RGB (MULTIMODAL GRAPH MODEL)

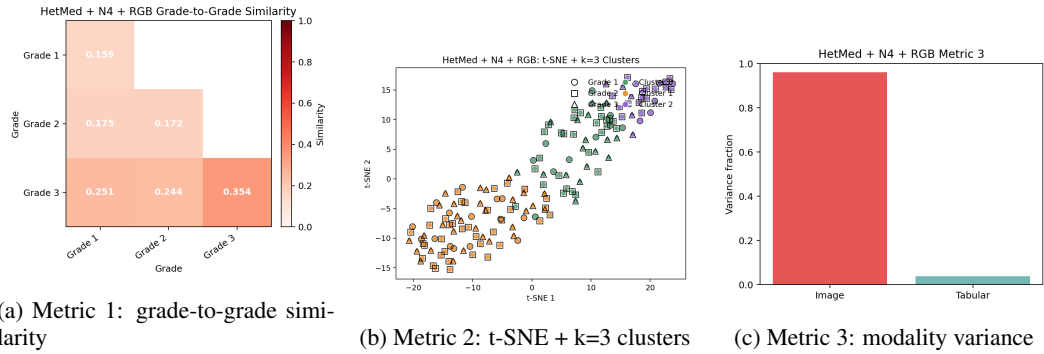


Figure 4: Intrinsic analysis for HetMed + N4 + RGB. We use the grade-to-grade similarity heatmap as the main visualization for Metric 1, a t-SNE cluster plot for Metric 2, and a variance plot for Metric 3. This subsection uses the improved rerun as the current main multimodal result and cleanly separates it from other model families that may be added later.

HETMED + RGB + SWIN (MULTIMODAL GRAPH MODEL)

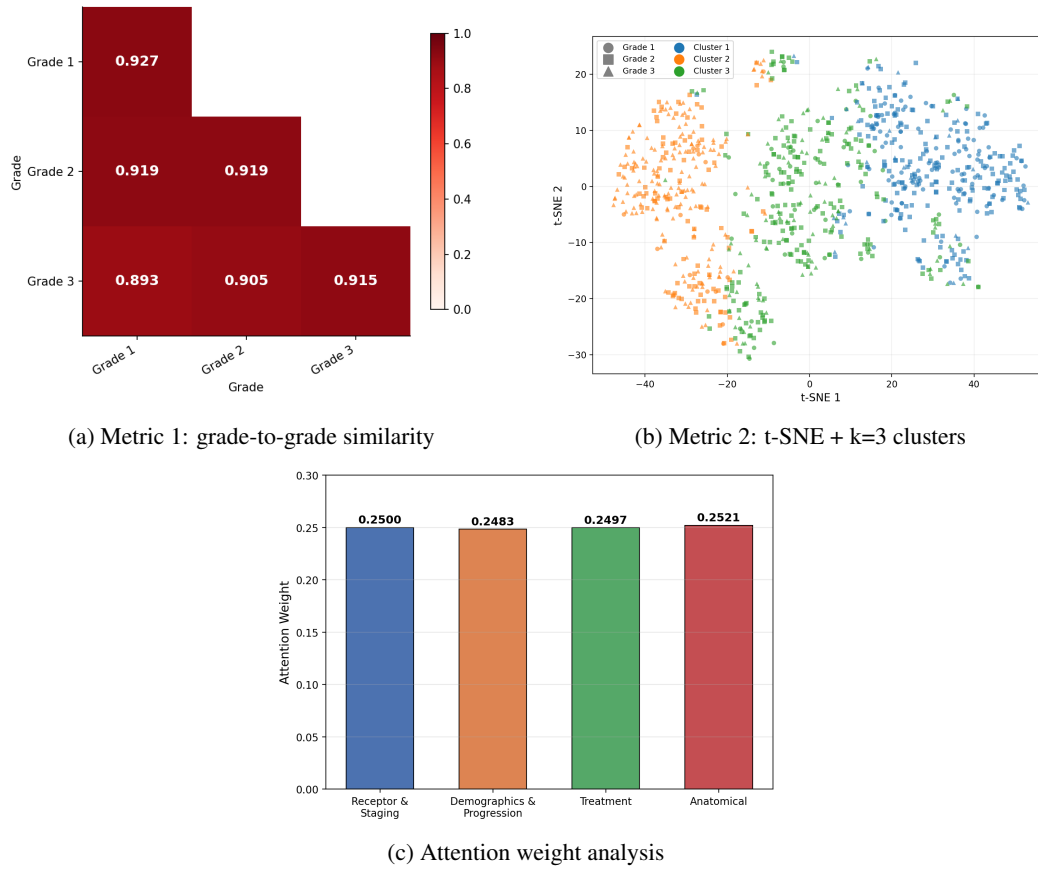


Figure 5: Intrinsic analysis for HetMed + RGB + Swin. The embeddings from HetMed framework lack discriminative power to classify tumor grades. This is because the model is not learning from the different relation types. As shown in (c), the model learns roughly equally from four types of features, diminishing the power of multiplex graph neural network.

HETMED + N4 + GRAYSCALE (MULTIMODAL GRAPH MODEL)

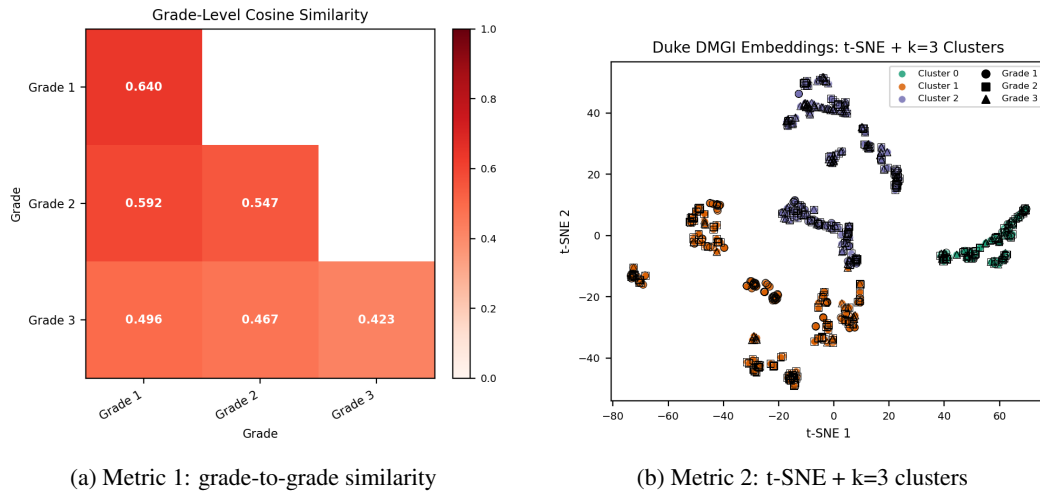


Figure 6: Intrinsic analysis for the HetMed+N4+Grayscale Multimodal Graph Model.

SIMCLR + GRAYSCALE (MULTIMODAL GRAPH MODEL)

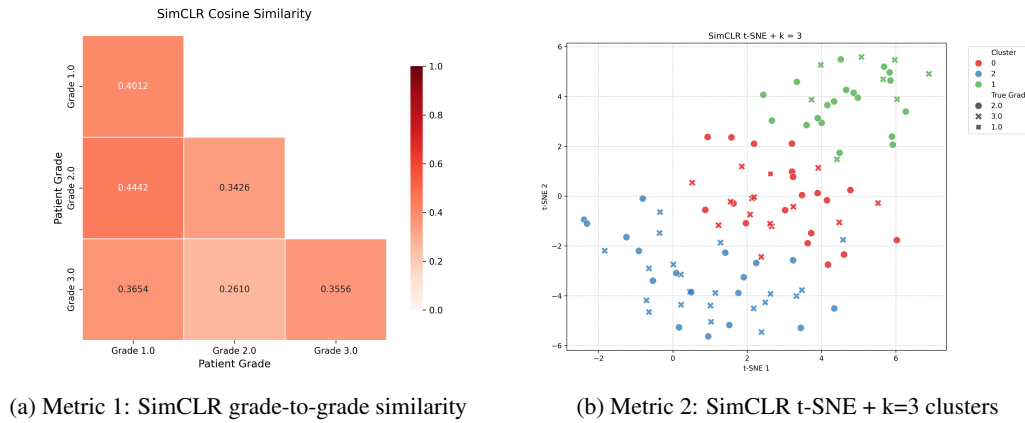


Figure 7: Intrinsic analysis for the SimCLR + Grayscale Multimodal Graph Model.

Methods	Dev Accuracy $\uparrow$
Unimodal 1: XGB (tabular only)	0.4649
Basic Multimodal MLP	0.5280
Basic Multimodal Swin + MLP	0.5027
SimCLR + Grayscale	0.4600
HetMed + RGB + Swin	0.2811
HetMed + N4 + Grayscale	0.5934
HetMed + N4 + RGB rerun	0.4573

Table 1: Task-level performance metrics for the methods currently available in this draft. Additional baselines and teammate variants can be filled in later.

Methods	Gap $\uparrow$	Dev		
		ARI $\uparrow$	NMI $\uparrow$	Purity $\uparrow$
Unimodal 1: XGB (tabular only)	0.0315	0.0325	0.0741	0.5189
Basic Multimodal MLP	0.0453	0.0298	0.0484	0.4972
Basic Multimodal Swin + MLP	0.6852	0.0517	0.0650	0.5385
SimCLR + Grayscale	0.00959	0.0066	0.0333	0.5900
HetMed + RGB + Swin	0.0139	0.0950	0.1172	0.5722
HetMed + N4 + Grayscale	0.0257	0.0761	0.0815	0.5228
HetMed + N4 + RGB	-0.0060	-0.0099	0.0101	0.4939

Table 2: Intrinsic clustering metrics for the current methods. Here, Gap denotes the difference between mean within-grade similarity and mean inter-grade similarity. Metric 3 is discussed in the text and visualized separately because it is more naturally interpreted through the modality-variance figure.

Methods	Dev	
	Image fraction	Tabular fraction
Basic Multimodal MLP	0.7524	0.2475
Basic Multimodal Swin + MLP	0.6752	0.3248
HetMed + RGB + Swin	0.6752	0.3248
HetMed + N4 + Grayscale	0.2976	0.7023
HetMed + N4 + RGB	0.9620	0.0380

Table 3: Metric 3 summary table for modality contribution. We report the variance fractions contributed by the image branch and the tabular branch before fusion.

1.2 QUALITATIVE ANALYSIS AND EXAMPLES (FULL PAGE TABLE)

[2 points] A full page table that includes: Example input, comparison of model outputs, theory about reason for failure. Every model presented should have at least one failure condition

Model-level qualitative analysis: XGB (tabular-only baseline). The XGB baseline shows that structured clinical variables alone already carry meaningful signal for Nottingham grade prediction. Its representation preserves a modest amount of grade-dependent organization under both similarity- and clustering-based analyses, which makes it a useful reference point for interpreting more complex models. At the same time, its qualitative limitation is clear: because it only consumes tabular inputs, it cannot use lesion morphology, enhancement pattern, or spatial context from MRI. As a result, its strengths lie in stable use of clinical covariates rather than in modeling visually rich disease structure.

Model-level qualitative analysis: SimCLR + Grayscale. The Resnet18-SimCLR + Grayscale framework successfully processes MRI data. It can be regarded as the baseline. Its internal representations struggle to clearly categorize the data by true grade.

Model-level qualitative analysis: HetMed + N4 + RGB. The HetMed + N4 + RGB pipeline illustrates both the promise and the current difficulty of multimodal fusion. The model can learn from processed MRI inputs together with non-image features, but the intrinsic results suggest that its embedding space is still only weakly organized by grade. The most informative clue comes from Metric 3, where the image branch contributes the overwhelming majority of pre-fusion variance. This suggests that the current limitation is less about whether the model can optimize at all, and more about whether the two modalities are being combined in a balanced and clinically meaningful way.

Model-level qualitative analysis: Basic Multimodal MLP This MLP was based on concatenating image embeddings produced by the Resnet-18 encoder with non-image features. The results from Basic MLP model show that simple fusion of tabular and image features can produce some grade-related signal, but the learned embedding space was still weakly organized by Nottingham grade. The variance analysis analysis for this test was particularly informative. Around 75.2 percent of the normalized variance comes from the image branch and only 24.8 percent from the non-image branch.

Model-level qualitative analysis: Basic Multimodal Swin + MLP The simple multimodal baseline shows that naive fusion of images and clinical features can generate discriminative embeddings where intra-grade pairwise cosine similarity is higher than inter-grade pairwise cosine similarity. However, the embeddings are not well-separated by tumor grades. In the K-Means clusters, each cluster contains embeddings of multiple grades and the clustering results are no better than guessing.

Model-level qualitative analysis: HetMed + N4 + Grayscale This version used N4 correction and intensity thresholding on grayscale images. The same grayscale image was replicated three times for each patient and used as input for the Densenet-121 encoder to produce image embeddings. When compared to the Basic Multimodal MLP strategy, this graph-based model produces a different qualitative pattern. Its embedding space shows slightly better class-related structure by clustering metrics and stronger within-grade coherence for some grades, especially grade 1. The modality-balance result is the opposite of the Basic MLP. Around 70.2 percent of the normalized variance comes from the non-image features and 29.8 percent is from the image features. This suggests that the GNN is more strongly shaped by the structured clinical graph construction and node attributes than by the MRI embedding component.

Model-level qualitative analysis: HetMed + RGB + Swin The model generates embeddings that are similar for all samples. This is because the model fails to learn from the various types of non-image features. This is shown by the attention weights analysis in Figure 5 where different types of clinical features have similar attention weights. It could be that the types of features are not discriminative in itself. Since the input of the model is concatenated embeddings of image and non-image embeddings projected into the same dimension, it also could be that information is lost in the project layers and the fused embeddings do not carry enough signal for graph-based learning.

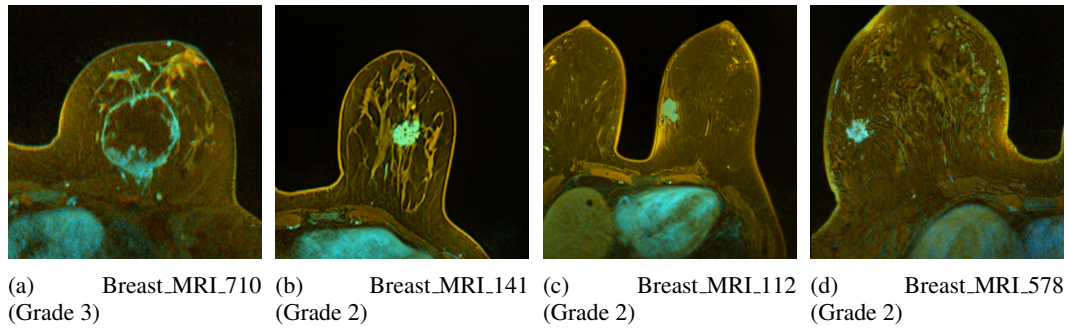


Figure 8: Representative processed MRI inputs for HetMed + N4 + RGB after channel-wise preprocessing and RGB fusion. These examples illustrate that the preprocessing pipeline produces visually usable anatomy and lesion contrast, but also highlight the diversity of lesion appearance across patients.

Patient ID	Grade	ER	PR	HER2	Staging N	Lymphadenopathy	Multicentric / Multifocal
Breast_MRI_710	3	Negative	Negative	Negative	1	Yes	No
Breast_MRI_141	2	Positive	Positive	Positive	0	No	Yes
Breast_MRI_112	2	Positive	Negative	Negative	0	No	No
Breast_MRI_578	2	Positive	Positive	Positive	2	Yes	Yes

Table 4: Example structured clinical inputs that are directly available to the XGB baseline. These rows are also useful for interpreting why XGB can remain competitive: even without MRI input, receptor status and nodal information already provide nontrivial grade-related signal. Additional columns or cases can be added later if the team wants a richer qualitative table.

Failure Theory - Basic Multimodal MLP and HetMed + N4 + Grayscale: The Basic MLP appears to underuse the non-image modality. This may be because the image branch dominates the shared embedding space and simple concatenation does not result in a balanced multimodal representation. The clinically relevant tabular data may be contributing less to the learned features as a result of the image variation in the basic multimodal MLP. The GNN learns better grade-related structure, but the embedding space is still not strongly separated by grade. Based on variance analysis, the non-image clinical data may be causing an imbalance in the learned embeddings. The GNN and MLP may be learning a useful signal, but each is imbalanced in a different way. As a result, the fused embeddings do not currently create clear distinctions between grade.

Example Inputs: To make the analysis more concrete, Figure 8 shows representative processed Comparison-3 MRI examples used by the HetMed + N4 + RGB pipeline, while Table 4 shows example structured clinical inputs of the kind directly consumed by XGB. Together, these examples make the two model inputs more tangible: HetMed receives processed RGB MRI slices plus tabular features, whereas XGB receives only the tabular clinical portion.

[3 points] Insights and insights Two paragraphs on what you found and how they impact your expectations for the main model, or how they have inspired you to change your proposed approach, etc

A central finding from the current experiments is that adding image features or graph structure does not automatically produce a better grade-aware representation. Some simpler baselines already retain meaningful grade-related structure, while several multimodal models still show weak clustering alignment or high cross-grade similarity despite using richer inputs. At the same time, the strongest multimodal results suggest that useful signal exists in both image and non-image modalities, but different architectures often over-emphasize one branch: some are image-dominated, while others are shaped much more strongly by tabular or graph features. Taken together, these results suggest that the central challenge of this project is not simply to build a larger multimodal model, but to learn a representation in which the two modalities contribute in a balanced and clinically meaningful way.

These findings change our proposal in two ways. First, we should treat balanced multimodal fusion as the main design target, rather than assuming that graph reasoning, stronger image encoders, or longer training alone will solve the problem. Second, future model selection should be guided by both extrinsic and intrinsic criteria: we should prefer models that not only improve accuracy, but also show cleaner within-grade similarity, better cluster-label alignment, and healthier modality balance.

If we continue building on the HetMed family, the most promising next adjustment is likely to be better control of how image and tabular features are projected and scaled before fusion, so that one branch does not dominate the consensus embedding. In practice, this means prioritizing feature projection, branch rebalancing, and more informative relation construction, while using the current simpler baselines as reference points for whether a proposed multimodal model is actually learning better structure rather than only optimizing better.

REFERENCES

A APPENDIX